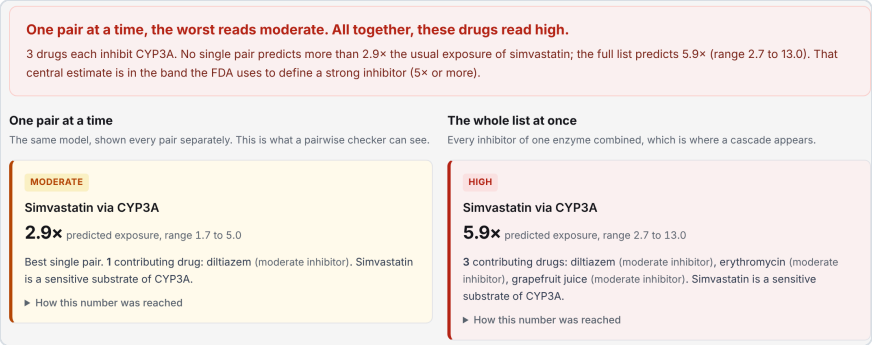
Cascade

The drug interaction a pairwise check cannot see.

Most interaction checkers compare drugs two at a time. A patient on seven drugs has 21 pairs. Three of those drugs can each partly inhibit the same enzyme, and their combined effect appears in no single pair, so a check that looks one pair at a time understates it. Cascade scores the whole list at once, with a model built only from the FDA's published interaction classes, and shows the pairwise view beside it. It takes the list the way people actually write it: brand names, misspellings and pharmacy printouts are read by a small trained model on the Cascade server, with no API key and no outside service.



Simvastatin, diltiazem, erythromycin, grapefruit juice, metoprolol, omeprazole and warfarin. Best single pair: 2.9× predicted exposure, moderate. The three CYP3A inhibitors together: 5.9× (range 2.7 to 13), high. Drop any one of them and the high reading disappears.

Across the FDA's index table of 205 drugs

Combination	Count	Understated by a pairwise view
2 inhibitors of one enzyme + 1 substrate	84,302	8,057 (9.6%)
3 inhibitors of one enzyme + 1 substrate	1,341,070	78,753 (5.9%)

"Understated": the full list lands in a higher severity band than any single pair of the same drugs. Of random 7-drug lists from the same table, 3.1% hold at least one (62 of 2,000).

Measured

Every 2- and 3-inhibitor combination in the table	1.4 million enumerated
Calibration: one inhibitor reads back in its own FDA class	holds for all three
Multiplying inhibitions instead, three moderate inhibitors	125× against 13× competitive
Reading 80 blind medication lists, table drugs	precision 99.1%, recall 95.2% (exact lookup: 50%)
Look-alike pairs from ISMP's confused-names list	never read as the partner
Tests	62 passing

How it works

1. **Read**: a neural network trained on realistic misspellings of drug names (RxNorm brands, the 300 most prescribed US drugs) turns a pasted list into table drugs. Common English words are never fuzzy-matched ("program" is not Prograf), and every reading is shown.
2. **Roles**: 285 published roles for 205 drugs, taken from the FDA's index table. A medicine the reader knows is outside it is reported as not covered.
3. **Intervals**: every FDA class is a range (a moderate inhibitor raises exposure 2 to 5×), and the ranges are carried through the arithmetic. The fraction metabolised is derived from the class definitions, not assumed.
4. **Combine**: inhibitors of one enzyme add competitively, $A = 1/(1+\Sigma r)$, $AUCR = 1/(fm \cdot A + 1 - fm)$.
5. **Compare**: the baseline is the same model run one pair at a time, so the only difference between the two views is the cascade.

Limits, stated up front

- It predicts exposure, not harm. Dose, kidney and liver function, genotype and timing are not modelled.
- One enzyme at a time, reversible inhibition only, and only the 205 drugs on the FDA's index list.
- The reader can misread (2 wrong drugs and 11 missed on the blind lists) and rarely asks when unsure. Every reading is shown.
- Not validated against clinical outcomes. It shows what a pairwise view misses, not that a missed cascade harmed anyone.

Educational prototype, not clinical decision support. Data: U.S. FDA, Examples of Drugs that Interact with CYP Enzymes and Transporter Systems, Tables 1 and 2 (retrieved 31 Oct 2026). Names: RxNorm (NLM, Nov 2026); ClinCalc Top 300 of 2024; ISMP List of Confused Drug Names. Built by Joshua Anojulu with Claude (Anthropic), which wrote the code and trained the reader. Every number comes from evaluate.py, evaluate_reader.py and the tests.